ABSTRACT: Since muscle coactivation increases the stiffness and stability of a joint, greater coactivation is likely during faster than slower movements. Very few studies, though, have been conducted to verify this hypothesis. Moreover, a large number of studies have examined coactivation of muscles surrounding the knee joint whereas there are few reports on the elbow joint. The aim of this study was therefore to compare the antagonist activation of the elbow flexors and extensors during isokinetic concentric exercises and to investigate the influence of angular velocity on their activation. Twelve men participated in the study. The surface electromyographic signals (sEMG) were recorded from the biceps brachii (BB) and triceps brachii (TB) muscles during three maximal voluntary isometric contractions (MVC) of elbow flexors and extensors and a set of three maximal elbow flexions and extensions each at 15°, 30°, 60°, 120°, 180°, and 240°.s^{-1}. Normalized root mean square (RMS) of sEMG was calculated during the isokinetic phase of movement as an index of sEMG amplitude. During elbow flexion, the antagonist activation of BB averaged 16.2% lower than TB, and this difference was statistically significant at all angular velocities. The normalized RMS values ranged from 26.0% ± 19.0 at MVC to 37.8% ± 13.9 at 240°.s^{-1} for antagonist TB activation, and from 5.7% ± 5.2 at MVC to 18.9% ± 8.6 at 240°.s^{-1} for antagonist BB activation. No influence of angular velocity on agonist and antagonist activity was found. Moreover, flexion and extension torques were both strongly affected by the amount of antagonist activation. The functional specialization of the two muscle groups could be responsible for the different levels of antagonist activation. The frequent use of BB, which is not assisted by gravity during daily activities, could lead to reduced coactivation due to a better functioning of the control system based upon reciprocal innervation. These findings may have significant implications in the design of rehabilitation programs directed to the elbow joint.

Muscle Nerve **33:** 191–199, 2006

COACTIVATION OF THE ELBOW ANTAGONIST MUSCLES IS NOT AFFECTED BY THE SPEED OF MOVEMENT IN ISOKINETIC EXERCISE

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Accepted 14 September 2005

The simultaneous activation of agonist and antagonist muscles acting on the same joint ("muscle coactivation") seems wasteful as their reciprocal action would impair the resultant muscle torque. The frequent use of coactivation, however, suggests that specific advantages exist for this strategy. Coactivation, in fact, represents a useful strategy when novel tasks are learned or a movement is performed requiring a high degree of accuracy.[5,13] Another advantage is that coactivation makes a joint stiffer and more difficult to perturb.[3,25] Joint stiffness, in turn, is influenced by the velocity of movement, being higher during faster movements.[8] Because coactivation increases the stiffness and stability of a joint, greater coactivation level is more likely during fast than slower movements. Very few studies, though, have been conducted to verify this hypothesis and

Abbreviations: BB, biceps brachii; CC, cross-correlation; DD1/DD2, double differentials; EE, elbow extension; EF, elbow flexion; EMG, electromyographic; MVC, maximal voluntary contraction; RMS, root mean square; RMS_{Ago}(%), percent of agonist RMS amplitude; RMS_{Ant}(%), percent of antagonist RMS amplitude; sEMG, surface electromyography; TB, triceps brachii

Key words: elbow extension; elbow flexion; isokinetic exercise; muscle coactivation; muscle cocontraction; sEMG; surface electromyogram

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Published online 23 November 2005 in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/mus.20462

there is no clear evidence in support of it. Certain previous studies on the knee joint found no differences in antagonist activity during isokinetic concentric exercises from 100° to 400° per second,[32] whereas others found higher levels of coactivation at faster speeds.[17,22] Moreover, coactivation reportedly varies over the range of movement, decreasing initially to allow faster acceleration and increasing at the end of the range to provide dynamic braking.[17] Taken together, these data suggest that when coactivation is evaluated without excluding the transient acceleration and deceleration phases, the influence of velocity cannot be isolated.

A large number of studies have examined coactivation of muscles acting around the knee joint.[1,3,11,19,21,34] During isokinetic knee extension, remarkably high antagonist hamstring activity (which may reduce the anterior-directed shear of the tibia) has been found.[1] During knee flexion, by contrast, Osternig and colleagues found only little antagonist coactivation of the quadriceps and suggested that the hamstrings could be used to a much greater extent than quadriceps for limb deceleration.[32,33] Surprisingly, coactivation of muscles surrounding other joints, such as the elbow, has been poorly investigated in comparison to the knee, and, to our knowledge, there are no reports on the differences between antagonist activation of elbow extensors and flexors.

The aims of this study were therefore (1) to verify whether the amount of antagonist activation of biceps brachii (BB) and triceps brachii (TB) is influenced by the speed of movement when transient phases of nonisokinetic activity are excluded from the analysis, and (2) to investigate whether there is any difference in the amount of antagonist activation between BB and TB during elbow flexion and extension.

MATERIALS AND METHODS

Subjects. With ethics committee approval, 12 young healthy men (age, 25.8 ± 4.4 years; height, 175.2 ± 9.5 cm; weight, 67.5 ± 13.2 kg) volunteered for this study and provided written informed consent. All individuals were physically active, but they were not involved in any kind of systematic training of the elbow flexors/extensors. Since the thickness of the subcutaneous layers has been shown to decrease electromyographic (EMG) signal amplitude[14] and to enhance crosstalk effects,[39] their skinfold thickness was measured by a skinfold caliper (Gima Plicometro; Gima, Milan, Italy) at the electrode location area on the BB and TB muscles (BB: 3.5 ± 0.3 mm; TB: 7.0 ± 1.0 mm).

Procedures. The elbow flexion and extension torques of the dominant limb were measured with an isokinetic dynamometer (Kin-Com, Chattanooga, Tennessee). Participants were seated comfortably in the dynamometer chair and were stabilized by chest and waist straps. The upper arm was fixed in a position parallel to the trunk and the forearm halfway between pronation and supination. The wrist was secured in a padded cuff attached to the load cell. The center of rotation of the lever arm was aligned to the distal lateral epicondyle of the humerus.

The surface electromyographic (sEMG) signals from the short head of BB and the lateral head of TB of the dominant limb were recorded by means of a linear array of four electrodes (silver bars 5-mm long, 1-mm thick, 10-mm apart; LISiN, Turin, Italy). The optimal position and orientation of the electrodes were determined to be conveniently distant from the innervation zone and the tendon, as previously described.[7] In particular, prior to final electrode positioning, attention was taken to verify that there was clear propagation in one direction of the action potentials without change in shape. Moreover, the cross-correlation (CC) function between the two double-differential signals (see below) was calculated. If the CC was >0.8, it can be assumed that there was a correct propagation of the action potentials and that the array was conveniently distant to the innervation zone. After gentle skin abrasion and cleaning with ethyl alcohol, electrodes were attached on the skin over the BB along a line connecting the acromion to the cubital fossa and over the muscle belly of TB along a line connecting the acromion to the olecranon. A ground electrode was placed around the wrist of the contralateral limb.

Surface EMG signals were detected in a single-differential mode. Double differentials were computed off-line in order to enhance the spatial resolution of the sEMG recording and, thus, to reduce crosstalk effects,[10,15,40] and were used for further analysis. Signals were amplified (×1,000), band-pass filtered (10 Hz–450 Hz; LISiN EMG 16), and sampled at 2,048 Hz. Signals were then A/D converted with 12-bit resolution (amplitude range ± 10 V; DAQ card AI-16XE-50, National Instruments, Austin, Texas), recorded, and stored on a personal computer for further analysis.

Experimental Procedure. Each individual visited the laboratory on two occasions. During the first visit, subjects were familiarized with all the procedures

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and exercises. During the subsequent experimental session, participants were requested to perform several tasks: (1) maximal voluntary isometric contractions (MVC); (2) maximal isokinetic concentric contractions at 15°, 30°, 60°, 120°, 180°, and 240°.s^{-1}; and (3) maximal isokinetic eccentric contraction at 15°.s^{-1} (required for normalization).

MVC. After a practice and warm-up trial at submaximal intensity, participants were required to produce three MVCs. The joint angle was fixed at 90° (isometric test). The MVC task consisted of rapidly increasing the force exerted by elbow flexors or extensors to a maximum. A target line was set on the computer screen at a value 20% higher than the best MVC, excluding the first trial. Subjects followed their performance on the computer screen and were verbally encouraged to achieve a maximum, in an attempt to exceed the target force,[4] and to maintain it for at least 2–3 s before relaxing. A minimum of three maximal attempts were performed, for both extension and flexion, separated by 5 min to recover from fatigue, and the highest performance was chosen for further analysis. Participants were asked to make further attempts if the MVC of their last trial exceeded by at least 10% that of previous trials.

Isokinetic Concentric Contractions. After the MVC task, participants were requested to perform a set of three maximal concentric elbow flexions and extensions each at 15°, 30°, 60°, 120°, 180°, and 240°.s^{-1}. The range of motion was 80° starting from 50° to 130° (90° horizontal position of the forearm). The order of the trials was randomized in order to minimize the effect of skill acquisition. Each contraction was followed by a rest period lasting from 5 min (after 240°.s^{-1}) to 30 min (after 15°.s^{-1}) to prevent cumulative fatigue.[29]

Isokinetic Eccentric Contractions. Subjects were also asked to perform a set of three maximal eccentric elbow flexion and extension at 15°.s^{-1}. The sEMG amplitude obtained during this task was used to normalize data of antagonist activation obtained during the concentric tasks.

Data Analysis. All data collected during the experiments were analyzed off-line (LabView 6.0.2 Software; National Instruments). The MVC that showed the highest value for force was chosen for the analysis. For the isokinetic task, the best trial of the set of three (i.e., the trial which showed the highest value of force) was preferred for the analysis. The root mean square (RMS) of sEMG was computed on the two double-differential signals (DD1 and DD2) over adjacent epochs of 125 ms overlapped by 62.5 ms. The double differential that showed the highest

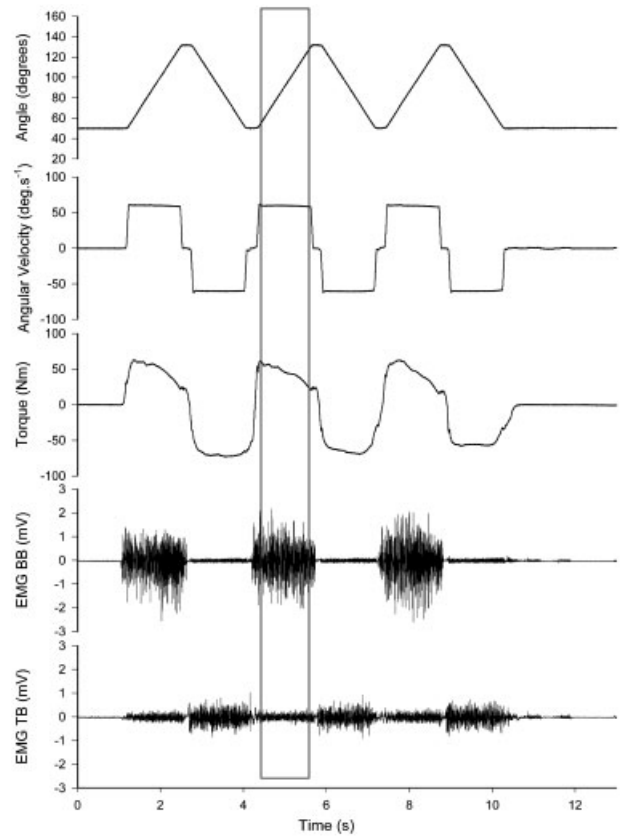

FIGURE 1. Angle, angular velocity, torque, BB and TB EMG signals recorded during a set of three concentric contractions at 60°.s^{-1}. The signals in the white rectangle were isolated and used for the subsequent analysis since they corresponded to the very isokinetic phase of the contraction (constant angular velocity).

RMS for the agonists was preferred to the other. For the MVC task, data were calculated on a 1-s window centered at the peak force. For the isokinetic contraction, data were analyzed over the period in which the velocity was constant (Fig. 1), in order to avoid the transient phases of the contractions (initial and final nonisokinetic activity).

Agonist RMS amplitudes were calculated as follows:

$$RMS_{Ago}(\%) = \frac{RMS_J}{RMS_{15CONC}} \times 100 \qquad (1)$$

where RMS_J is the average RMS value at a given angular velocity (j = 15 °, 30°, 60°, 120°, 180°, and 240°.s^{-1}); and RMS_{15CONC} is the average RMS value at 15°.s^{-1} concentric contraction.

Antagonist RMS amplitudes were calculated as:

$$RMS_{Ant}(\%) = \frac{RMS_J}{RMS_{15ECC}} \times 100 \qquad (2)$$

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where RMS_{15ECC} is the average RMS value at 15°.s^{-1} eccentric contraction.

Following Aagaard and colleagues,[1] this normalization was adopted in order to be consistent with the fact that during any antagonist activation the muscle is lengthened instead of shortened, a condition that favors higher torque and lower EMG activation.[22] For the same reason, agonist RMS amplitudes were calculated with respect to the average RMS value of the maximal concentric contraction at 15°.s^{-1}.

As the peak torque is attained earlier from slower to faster isokinetic contractions (hence at different angles)[20] and as the linearity of EMG–force relationship is affected by muscle length and joint angle,[6] we preferred to calculate the average RMS value over the entire (isokinetic) range of motion as suggested by Schmitz and Westwood.[37]

Statistical Analysis. A repeated-measures analysis of variance [between factors: muscle (BB vs. TB); within factors: modality of activation (agonist vs. antagonist), and velocity of movement (0°, 15°, 30°, 60°, 120°, 180°, and 240°.s^{-1})] was used to compare the dependent variables (RMS and force). A *t*-test with Bonferroni correction was implemented when appropriate. Data are expressed as mean ± SD in the text and mean ± SE in figures and tables. Statistical significance was accepted if the *P*-value was <0.05. Regression lines for individual data sets of $RMS_{Ago}(\%)$ and $RMS_{Ant}(\%)$ versus torque were computed using the least-squares method.

RESULTS

Torque. The torque/velocity relationships for elbow extensors (EE) and elbow flexors (EF) are shown in Figure 2. An exponential regression line was used to fit the averaged points and the regression coefficients were 0.97 for EE and 0.98 for EF. Torque values ranged from 70.5 ± 22.7 Nm at MVC to 32.3 ± 12.5 at 240°.s^{-1} for EF and from 79.1 ± 32.1 Nm at MVC to 42.4 ± 15.0 at 240°.s^{-1} for EE. Despite the fact that the EE torque/velocity curve was above the EF curve, no statistical difference was found between the two data sets.

Figure 3 shows the relationships between maximal torque and EMG activation. In particular, EE and EF torques were inversely correlated to $RMS_{Ant}(\%)$ for both BB (r^2 = 0.96) and TB (r^2 = 0.93), respectively. These correlations were much higher than those for $RMS_{Ago}(\%)$ and torque both in EF (r^2 = 0.50) and EE (r^2 = 0.63).

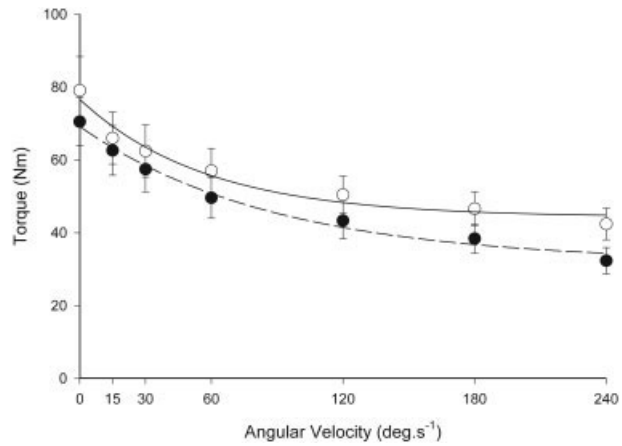

FIGURE 2. Torque/angular velocity relationships for EF (filled circles) and EE (open circles). Exponential regressions have the form Y = 44.33 + 32.36e$^{-0.017X}$ for EE (continuous line) and Y = 31.93 + 37.26e$^{-0.011X}$ for EF (dashed line). Data are reported as mean ± SE.

sEMG Activation. ***Skinfold Thickness and RMS.*** In Figure 4, the individual level of BB and TB antagonist and agonist activation, obtained during the MVC task, are plotted as a function of the skinfold thickness. The data distribution of both graphs is such that it was not possible to detect any significant correlation between skinfold thickness and level of coactivation, and between skinfold thickness and agonist activation.

Agonist Activation. $RMS_{Ant}(\%)$ for BB ranged from 120.9 ± 27.9 during MVC to 93.1 ± 16.0 at 240°.s^{-1}. Similar behavior was shown by $RMS_{Ago}(\%)$ for TB, ranging from 105.6 ± 25.5 during MVC to 94.5 ± 18.2 at 240°.s^{-1} (Fig. 5 and Table 1). No statistical differences were found between the two muscle groups and among the different angular velocities.

Antagonist Activation. $RMS_{Ant}(\%)$ for BB ranged from 5.7 ± 5.2 during MVC to 18.9 ± 8.6 at 240°.s^{-1}. TB showed a $RMS_{Ant}(\%)$ ranging from 26.0 ± 19.0 during MVC to 37.8 ± 13.9 at 240°.s^{-1} (Fig. 5 and Table 1). Statistical differences were present between the two muscles at each angular velocity ($P <$ 0.01). Moreover, for BB, there was a statistical difference between the $RMS_{Ant}(\%)$ at 60°.s^{-1} and at 120°.s^{-1} ($P < 0.01$).

DISCUSSION

We found that antagonist activation of BB was, on average, 16.2% lower than TB antagonist activation, and this difference was statistically significant at each of the examined angular velocities. The greater coactivation of TB during EF could be

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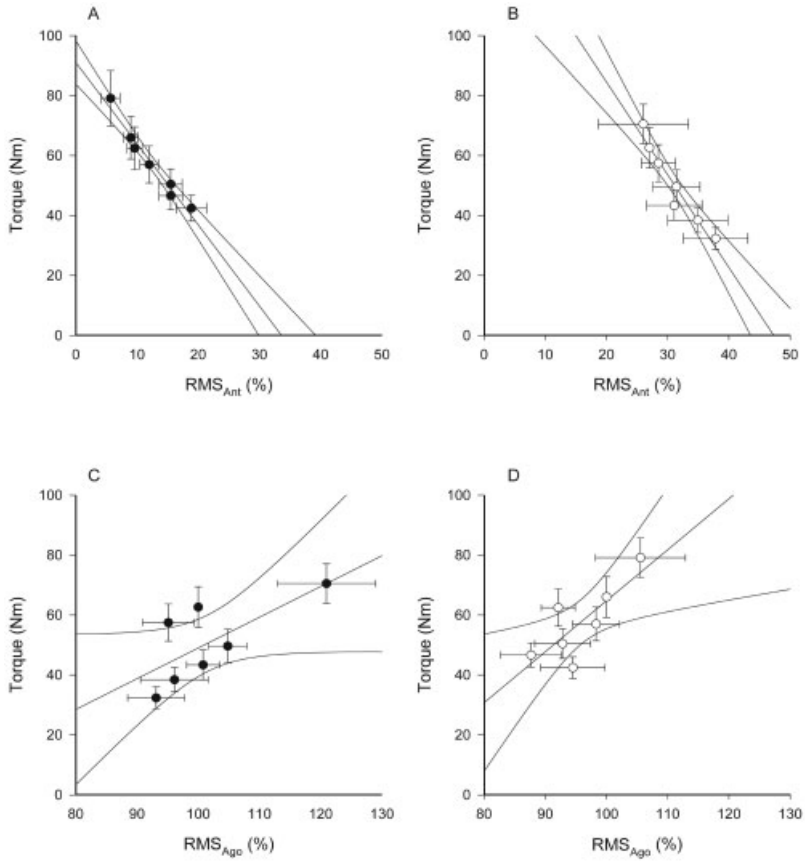

FIGURE 3. Maximal mean values of torque obtained at each angular velocity as a function of RMS_{Ago}(%) and RMS_{Ant}(%). **(A)** EE maximal torque versus BB RMS_{Ant}(%); **(B)** EF maximal torque versus TB RMS_{Ant}(%); **(C)** EF maximal torque versus BB RMS_{Ago}(%); **(D)** EE maximal torque versus TB RMS_{Ago}(%).

responsible for the tendency of the EF torque/velocity curve to be below the EE curve. Contrary to our expectation, statistical analysis showed that antagonist activation was not velocity-dependent for both muscles, even if BB showed a tendency to increase antagonist EMG activity as the angular velocity increased. These findings suggest that co-activation of the elbow extensors play the main role in the control of the forces generated around the elbow joint in order to guarantee the stiffness of the whole joint.

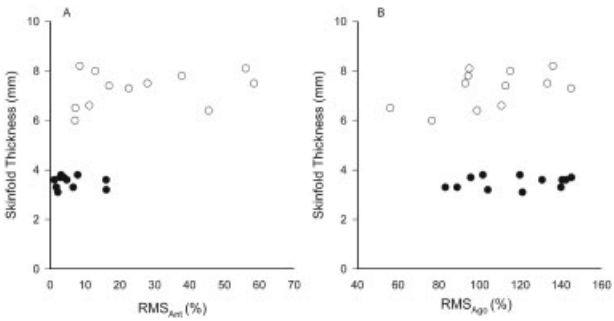

FIGURE 4. Scattergrams of BB (filled circles) and TB (open circles) skinfold thickness as a function of antagonist **(A)** and agonist **(B)** activation during MVC.

Torque/Angular Velocity Relationship and RMS. Elbow extensors and flexors showed a similar relationship between torque and angular velocity (Fig. 2). In fact, mean data of the two curves were not statistically different even if the EE curve was slightly higher than the EF one. Possible explanations for this trend could be that the total physiological cross-sectional area of TB is greater than BB. This is, however, counterbalanced by a smaller moment arm of the former with respect to the latter.[30] Accordingly, the ratio between the torques exerted by these two muscular groups has been reported to approximate unity.[23] Hence, the trend observed in this study could be explained by the greater antagonist activation of TB with respect to BB, which could have lowered the net flexion torque. This is in agreement with the high correlation found between $RMS_{Ago}(\%)$

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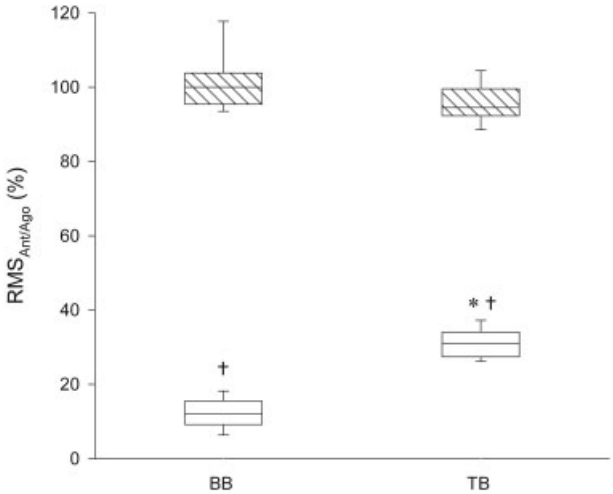

FIGURE 5. RMS_{Ago}(%) and RMS_{Ant}(%) for BB and TB when activated as agonist (hatched box) and antagonist (white box). Data are mean ± SE. Significant differences between muscles (*) and between agonist and antagonist activation (†) ($P < 0.05$).

and torque (Fig. 3). Moreover, this hypothesis is supported by studies that revealed that antagonist activity generates a constant opposing torque throughout isokinetic movement.[3,38] Additional evidence of the role of coactivation in the force-generation capacity is the age-related decline in force production observed in 70-year-old compared to 20-year-old individuals, which is accompanied by higher levels of antagonist activation.[27]

Coactivation of Agonist and Antagonist Muscles. Agonist activation of these two muscles was similar and was always close to the maximum value chosen for the normalization (100% RMS). This is evidence that subjects were successful in producing maximal-effort contractions even at high angular velocities, as they were requested. However, antagonist activation of TB was greater than BB at all angular velocities (Table 1). Contrary to our expectations, velocity of movement did not significantly enhance the antagonist activation of TB and BB. There are conflicting findings on the effect of angular velocity on antagonist activity.[21] Some authors reported a similar antagonist activation of ankle and knee extensors/flexors muscles during several isokinetic exercises performed at different speeds.[32,18] Kellis and Baltzopoulos,[22] by contrast, reported significant increases of antagonist hamstrings activity as the angular velocity increased. Similar results were found by Hagood and colleagues,[17] who reported a 128% increase in EMG activity of the hamstrings during concentric knee extensions from 15° to 240°.s^{-1}. This inconsistency of results could be explained on the basis of the different methods used in data analysis. In fact, Hagood and colleagues[17] included in their analysis the initial and final phases of movement and found a significant increase (>100%) in antagonist coactivation during the final 40° of fast extension/flexion and a minor decrease (8% for quadriceps, 24% for hamstrings) during the initial phase of movements. In the present study, we failed to find a significant relationship between antagonist activation and angular velocity, perhaps because we did not include nonisokinetic phases in the analysis of our data. Since coactivation of antagonist muscles occurs mainly during the acceleration and deceleration phases of movements and since these phases are longer at faster speeds, this is likely the reason for the increased coactivation reported by other studies at the higher speeds. Nevertheless, a slight tendency for a higher antagonist activation at higher speeds was observed in both muscles, and was more pronounced in the BB. As velocity of movement increased, more stability and stiffness are probably required to protect the joint, but, again, this phenomenon is more relevant during the initial and final phases of movement. This implies that more than velocity, antagonist activation is influenced by acceleration and deceleration actions. Moreover,

Table 1. RMS_{Ago} (%) and RMS_{Ant} (%) as a function of angular velocity.

Angular velocity (deg.s^{-1})	RMS_{Ago} (%)		RMS_{Ant} (%)	
	BB	TB	BB	TB
0 (MVC)	120.9 ± 8.0	105.6 ± 7.3	5.7 ± 1.5*	26.0 ± 5.5
15	100	100	9.0 ± 1.2*	27.0 ± 5.0
30	95.1 ± 4.2	92.1 ± 2.8	9.6 ± 1.2*	28.5 ± 4.5
60	104.8 ± 3.1	98.3 ± 3.8	12.0 ± 1.5*,†	31.4 ± 4.5
120	100.8 ± 2.7	92.8 ± 4.5	15.5 ± 1.8*	31.1 ± 3.5
180	96.1 ± 5.5	87.7 ± 4.9	15.6 ± 1.9*	34.9 ± 2.8
240	93.1 ± 4.6	94.5 ± 5.3	18.9 ± 2.5*	37.8 ± 4.0

Ago, agonist; Ant, antagonist; BB, biceps brachii; MVC, maximal voluntary contractions; RMS, root mean square; TB, triceps brachii.
Data are mean ± SE. Significant differences are shown (P < 0.05) between muscles (*) and between a given angular velocity and the subsequent one (†).

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Hagood and colleagues[17] normalized the EMG activity angle by angle throughout the range of motion. In the present study, the average RMS was calculated over the entire (isokinetic) range of movement in order to allay the concerns of a nonlinear EMG–force relationship during isokinetic movements[6] and of the earlier appearance of peak torque during slow isokinetic contractions with respect to faster ones.[20]

Differences in antagonist activation have been reported previously between quadriceps and hamstrings muscles.[1,34] In particular, the greater coactivation of hamstrings during active knee extension compared to quadriceps during knee flexion has been suggested to counteract the high anterior shear forces elicited when the quadriceps is maximally activated, a risk that is not present in the opposite direction.[21] In the elbow joint, similar destabilizing forces and displacement of forearm bones may involve a different biomechanical solution that includes, for example, the "mechanical stop" due to the olecranon. During elbow extension, in fact, we failed to find such a protective activity of the BB, or better, the antagonist activation of the TB was much higher. This different behavior of the extensors/flexors muscles between the knee and elbow joints suggests that there are factors other than the extensor/flexor activity to regulate forces and stiffness of the elbow joint.

Our data indicate that there is a correspondence between BB and quadriceps activities and between TB and hamstrings activities, but the complexity of the agonist/antagonist control mechanism makes it difficult to draw conclusions. It can be speculated that gravitational forces involved in the movement influence agonist/antagonist activity. In fact, when the leg is extended from a sitting position, the direction of the movement is against gravity and the knee flexors are mainly coactivated as antagonists. When the forearm is flexed starting from the position of the upper limb parallel to the trunk, the direction of the movement is against gravity as well, and the elbow extensors are the antagonists.

Changes in maximal neuronal activation and strength may vary among different muscles in relation to their use in daily physical activities and training,[12] and changes in coactivation of the antagonists may take place as well.[9] This could explain the differences between TB and BB muscles. A less frequent concentric activation of the TB, which is assisted by gravity during most daily activities, could be responsible for a different specialization of this muscle with respect to the BB. This lack of "training" could be the reason for the greater TB antagonist activation, and, on the other hand, the frequent use of BB could lead to a reduced coactivation due to a better functioning of the control system based upon the reciprocal innervation.

The problem in understanding the estimates of coactivation during movement is that tonic neural signals coexist with those associated with phasic muscle activity and activation elicited by reflexes. Several investigators have studied the effect of training and type of activity on muscle activation originating from Golgi tendon organs or from motor spindles,[31,35,36] and it has been suggested that tension-regulating processes mediated by reflexes are also trainable.[2] Future studies should be designed in a way to investigate the effect of training and type of activity on the coactivation of the elbow flexors/extensors and the role of the above-mentioned complex regulatory mechanisms on agonist/antagonist activation.

EMG Recording and Analysis. One of the most debated issues associated with the use of sEMG recording to examine coactivation is the detection of signals generated by the contraction of adjacent muscles. This phenomenon, known as crosstalk,[10] may lead to erroneous conclusions. Many studies have therefore focused on factors in data collection and analysis that can be modified to reduce crosstalk.[10,15,16,26,40] In this study, we adopted a "spatial filter" represented by the double-differential configuration. Common signals present in single bipolar recordings can be nullified by the second subtraction step.[10,24] Moreover, we choose a 10-mm interelectrode distance, which halves crosstalk compared to a 20-mm distance when associated with DD configurations.[15] To date, for optimal electrodes positioning, this kind of spatial filters represents the most effective approach to attenuate crosstalk.[40] Commonly used methods to quantify and reduce crosstalk, such as cross-correlation coefficient and high-pass temporal filtering, are not reliable.[16,26]

Care was also taken in identifying the location of the arrays on the muscle bellies. In fact, the amplitude of EMG signals detected close to the innervation zone could be reduced and lead to erroneous estimates of coactivation.[28] For this reason, prior to final electrode positioning, we verified that there was a clear propagation of the action potentials without any change in shape to ensure a convenient distance from the innervation zone.

Moreover, since excess subcutaneous adipose tissue can be responsible for greater crosstalk,[39] we measured BB and TB skinfold thickness and compared the values with sEMG amplitude. The poor correlation between this parameter and RMS can be interpreted as an index of a good detection of sig-

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nals, which were not altered by the filter represented by the adipose tissue. Thus, even if some contamination by signals generated from nonactive tissues cannot be excluded completely, it is unlikely that the sEMG in this study was significantly influenced by crosstalk.

Another important issue when studying coactivation is the normalization of signals, which has been shown to heavily influence data interpretation and conclusions. One of the most widely used methods is to express the EMG relative to the maximum isometric value.[22] This method, however, seems to be reliable only when isometric contractions are involved. Other authors have normalized antagonist activation relative to agonist EMG obtained during concentric contraction.[3,17] Hagood and colleagues, for example, normalized the antagonist EMG activity to the maximum activity obtained by the muscle during concentric isokinetic contraction at $15°.s^{-1}$.[17] During eccentric muscle activation (i.e., during antagonist activation), sEMG amplitude is lower compared to the concentric condition.[22] Thus, the normalization method based on signals recorded from concentric contraction may underestimate antagonist activity. In this study, we expressed antagonist activation as a percentage of the maximum value obtained during the isokinetic eccentric contraction at the lowest angular velocity ($15°.s^{-1}$) when the muscle acted as agonist. In this case, the normalization takes into consideration the muscle action (eccentric and isokinetic).

In conclusion, we have quantified the agonist and antagonist activation of BB and TB during maximal isokinetic concentric exercises performed at different angular velocities and found that the TB muscle is more activated than the BB when acting as antagonist at any of the presented velocities of movement. Although torque/velocity relationships of elbow flexors and extensors were not statistically different, the slightly smaller torques of the flexors were likely influenced by the greater antagonist activation of the TB as revealed by the high correlation between antagonist activation and agonist torque. Analogies with the knee joint are very difficult to make because of the complexity of the two joints and their regulatory mechanisms of control, but the gravitational forces involved in the knee extension and elbow flexion may have influenced agonist/antagonist activation more than the extensor/flexor specialization of the muscle groups surrounding the knee and the elbow. The antagonist activation of TB and BB muscles did not increase as the angular velocity increased, as reported by other authors for the knee joint. This fact can be attributed to differences in the data analysis, which in our study was not performed on the nonisokinetic phases of movement. All these findings may have important implication for the proper design of rehabilitation programs directed to the elbow joint. In fact, stiffness of the elbow joint following trauma is common and caused by many factors such as contractures of the joint capsule, ligamentous structures, intra-articular adhesions, but also musculo-tendinous structures. Surface electromyography can represent a useful tool to assess the condition of these structures and to monitor their recovery process. Understanding the details of elbow muscles' activation during different dynamic movements is thus a key factor in rehabilitating the elbow joint successfully.

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